

EDUCATION

Dec 2026 Pittsburgh, PA	Carnegie Mellon University MS in Computer Science - Machine Learning & Natural Language Processing (GPA: 4.17/4.0) Coursework: Advanced Natural Language Processing, Generative AI, Multimodal ML, LLM Systems Teaching Assistant: Advanced Natural Language Processing (Spring 2026)
June 2025 Pilani, India	Birla Institute of Technology and Science, Pilani BE in Computer Science (CGPA: 9.91/10 , Institute Silver Medalist - Rank 2)

PUBLICATIONS

ICCV'25, EACL'26	CAIRE: Cultural Attribution of Images by Retrieval-Augmented Evaluation. CEGIS @ ICCV'25, EACL'26 (Main Conference) Paper Code
LREC-COLING'24	BERT-based Idiom Identification using Language Translation and Word Cohesion. Multiword Expressions and Universal Dependencies @ LREC-COLING'24 Paper Code
ICML (under review)	GameDevBench: Evaluating Agentic Capabilities Through Game Development. International Conference on Machine Learning (ICML 2026)

EXPERIENCE

Carnegie Mellon University, Machine Learning Department Graduate Student Researcher	Pittsburgh, PA Aug 2025 – Present
- Developing GameDevBench, a scalable benchmark for evaluating multimodal LLM and computer-use agents (CUAs) in agentic Godot game development, comprising ~200 tutorial-derived tasks with automated task and test generation. - Built automated task-quality scoring using pixel-level metadata and VLM-judge-assessment, eliminating manual validation. - Under review at ICML 2026.	
Carnegie Mellon University, Language Technologies Institute Research Intern (Undergraduate Thesis), NeuLab Advisor: Prof. Graham Neubig Code	Pittsburgh, PA May 2024 – Mar 2025
- Built CAIRE, a retrieval-augmented evaluation system for cultural attribution in images, grounding visual content via large-scale entity linking. Implemented efficient retrieval over a 6M-entity FAISS index with multimodal SigLIP embeddings, outperforming LVLM baselines on fine-grained object grounding (FOCI benchmark). - Improved visual entity linking precision by reranking retrieved candidates using text-based semantic disambiguation. - Showed that CAIRE enables open-source VLMs to outperform frontier models on cultural relevance evaluation by conditioning predictions on retrieved cultural context, achieving +28% F1 and Pearson $r > 0.65$ alignment with human judgments; accepted at ICCV-W and EACL (Main Conference).	
Amazon, Applied Science Summer Intern Advisor: Jitenkumar Rana	Bangalore, India May 2023 – Aug 2023
- Product entity extraction (NER): fine-tuned a BERT-based token classification model to extract brand and model fields from noisy product webpages, producing structured entities for downstream product knowledge bases. - Shipping cost anomaly detection: trained regression models to estimate expected shipping costs beyond a rule-based heuristic, flagging anomalies via prediction residuals and reducing false negatives by ~25%; deployed via a Django REST API.	
Nanyang Technological University Research Intern, SpeechLab Advisor: Prof. Chng Eng Siong Code	Singapore Mar 2024 – Sep 2024
Built a text-based depression detection system by LoRA-fine-tuning LLaMA-3.1-8B on DAIC-WOZ, reformulating prediction as PHQ-8-aligned symptom scoring for interpretability and structured reasoning; leveraged transcript preprocessing and LLM-based synthetic dialogue augmentation, achieving +7.1% F1 over prior text-only baselines.	

PROJECTS

Hybrid Retrieval RAG System with Qwen2.5
- Built a Qwen2.5-7B RAG system using hybrid retrieval (MXBAI dense + BM25 sparse) with RRF. - Implemented grid-search evaluation over retrieval hyperparameters using accuracy, BLEU, BERTScore, and LLM-as-Judge.
Structured Agentic Reasoning with Diffusion Language Models
Fine-tuned diffusion language models (Fast-dLLM v2, 1.5B) to act as ReAct agents, generating structured Thought–Action–Observation trajectories and improving tool-call reliability (5% → 60%) while reducing trajectory length (9.2 → 6.4 steps).

SKILLS

Programming & OS: Python, C/C++, Java, SQL, Linux, Git, REST APIs, High Performance Computing Clusters (HPC)
Libraries & Frameworks: PyTorch, Scikit-Learn, HuggingFace, PEFT (LoRA), FAISS, Django, NumPy, Pandas